

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457697.6371 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.131 +/- 0.042
Transit depth (Rp/Rs)^2	1.71 +/- 1.11 [%]
Orbital Inclination (inc)	84.06 +/- 2.51
Airmass coefficient 1 (a1)	1.022 +/- 0.022
Airmass coefficient 2 (a2)	-0.0119 +/- 0.0042
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	#3 - [278, 313]
Optimal Aperture	2.87
Optimal Annulus	13.65
Transit Duration (day)	0.054 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.131 ± 0.042 vs 0.1326 (deviation 0.04σ)
Duration fit vs published	0.054 d vs 0.0737 d (deviation 0.55σ)
Mid-time O–C	-6.5 min
Depth SNR	3.0
Residual scatter	2.11 %

Light curve

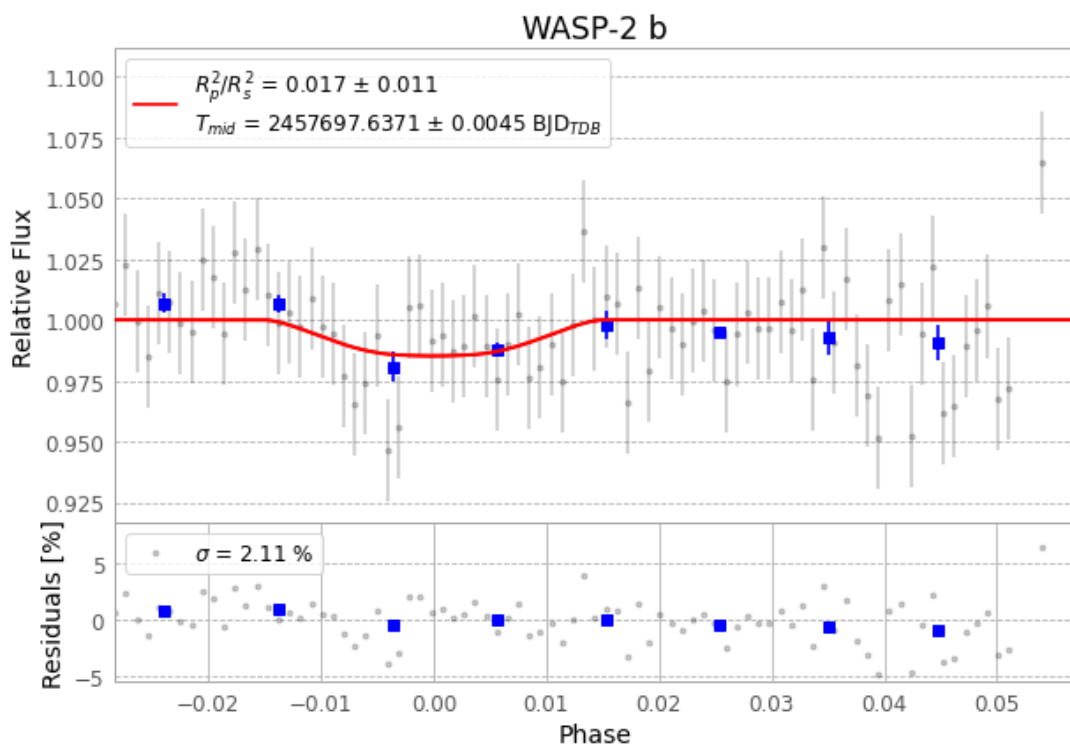


Figure 1 — FinalLightCurve_WASP-2 b_2016-11-04.png (SHA-256 bc5df5070a58ce83...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [472, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a70a26efd207ff30...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

90 FITS frames (59 MB), WASP-2161105014812.FITS → WASP-2161105061512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-161105.log (744290cf1468...), FinalLightCurve_WASP-2 b_2016-11-04.png (bc5df5070a58...), FinalLightCurve_WASP-2 b_2016-11-04.csv (9579dce6bfb0...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 01:38Z.